

The $(1 + t)$ is the last generator of each run:
a proof of the cylindric Newton identity in the
affine nil-Temperley–Lieb algebra

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Status. Theorem 6, Theorem 8 and Theorem 11 are *proved*. Proposition 13 is *proved*. The uniqueness half of gap (G1)(i) of [2] — that $t = -1$ is the *only* point at which $\mathcal{R}_e(t)$ commutes with the strip adders — remains open and is restated as (H1) in §7.

What this supersedes. This note closes gap (G1)(ii)–(iii) and gap (G3) of 2026-09-23-c2-ribbon-height-is-a-heap-orientation.tex ([2], commit clio-vega/proofs@0a01bb9), and makes gap (G2) of that note *moot*: the argument below is uniform in q and never separates the $q = 0$ and q^1 parts, so the “ $q = 0$ reduction” that (G2) proposed as a plan is not needed and is not carried out. Theorem thm:B(ii) and (iii) of [2] are hereby upgraded from *computed* to *proved*, and Proposition prop:MN of that note (the quantum Murnaghan–Nakayama rule) becomes unconditional.

Novelty is unchecked and no priority claim is made. A quantum Murnaghan–Nakayama rule for $\mathrm{QH}^*(\mathrm{Gr}_{kn})$ may well be in the literature; Bertram–Ciocan-Fontanine–Fulton’s quantum rim-hook rule and Morrison–Sottile’s *Murnaghan–Nakayama rules in Schubert calculus* are the obvious candidates and **I have not read either**. Nothing here is claimed to be first. What is claimed is that the proof below is a proof: it derives Theorem 11 from Postnikov’s quantum Pieri corollary and a two-line count in the trace monoid of $\mathcal{A}_n$.

Abstract

Let $\mathcal{A}_n$ be Postnikov’s affine nil-Temperley–Lieb algebra and let $\mathcal{R}_e(t) = \sum_{J,\varepsilon} t^{|\varepsilon|} a_{J,\varepsilon}$ be the height-graded cylindric e -ribbon adder, realised inside $\mathcal{A}_n[t]$ in [2]. We prove the cylindric splitting identity

$$\sum_{b=0}^e t^b \mathbf{h}_{e-b} \mathbf{e}_b = \sum_{|I|=e} \sum_{\varepsilon} t^{|\varepsilon|} (1+t)^{r(I)} a_{I,\varepsilon} \quad (1 \leq e \leq n-1),$$

where $r(I)$ is the number of cyclic runs of $I \subseteq \mathbb{Z}/n\mathbb{Z}$. The proof is two observations about heaps: a horizontal-strip monomial *empties* its own support while a vertical-strip monomial *requires* its support to be occupied, so the two supports must be disjoint; and once they are disjoint, the commutation class of the product is determined by which part each *edge-indexing* site lies in — leaving the last site of each run free. That free binary choice is the factor $(1+t)$. Differentiating at $t = -1$ kills every I with two or more runs and yields the closed formula $\mathcal{R}_e(-1) = \sum_{b \geq 1} (-1)^{b-1} b \mathbf{h}_{e-b} \mathbf{e}_b$, which exhibits $\mathcal{R}_e(-1)$ as a quantum multiplication operator and proves the cylindric Newton identity $r \mathbf{h}_r = \sum_{e \leq r} \mathcal{R}_e(-1) \mathbf{h}_{r-e}$ for $r \leq n-1$, together with its defect $(-1)^{k-1} (n-k)q$ at $r = n$. We also settle the boundary cases $k \in \{1, n-1\}$: there $\mathcal{R}_e(t)$ is a *monomial* in t , so no value of t is distinguished and the rigidity question is vacuous.

1 Conventions, imported verbatim from [2]

Throughout, $n \geq 2$ and $\mathcal{A}_n$ is the associative $\mathbb{Z}$ -algebra on generators a_i , $i \in \mathbb{Z}/n\mathbb{Z}$, with $a_i a_i = a_i a_{i+1} a_i = a_{i+1} a_i a_{i+1} = 0$ and $a_i a_j = a_j a_i$ for $i - j \not\equiv \pm 1$. It acts on $\bigoplus_k \text{QH}^*(\text{Gr}_{kn})$ by $a_i = E_{i,i+1}$ for $0 \leq i \leq n-2$ and $a_{n-1} = q E_{n-1,0}$, where E_{ij} moves a bead from site i to site j when i is occupied and j is empty and kills the basis vector otherwise. A partition $\lambda \in P_{kn}$ is the bead set $S = \{\lambda_j + k - j : 1 \leq j \leq k\} \subseteq \mathbb{Z}/n\mathbb{Z}$, $|S| = k$. Words are read in *application* order: the leftmost factor acts first ([2, **rem:conv**]; this is gap (G4) there and it is load-bearing here too, since every statement below is about application order).

Call $I \subseteq \mathbb{Z}/n\mathbb{Z}$ *proper* if $I \neq \mathbb{Z}/n\mathbb{Z}$. For proper I , the subgraph of the n -cycle induced on I is a disjoint union of paths; its maximal paths are the *runs* of I , of which there are $r(I)$, and $\text{Ed}(I)$ is its edge set, $|\text{Ed}(I)| = |I| - r(I)$. We index each edge by its smaller endpoint *inside its run*, so that

$$\text{Ed}(I) \longleftrightarrow I \setminus L(I), \quad L(I) := \{\text{last site of each run of } I\}, \quad |L(I)| = r(I), \quad (1)$$

the edge indexed by s being $\{s, s+1\}$. For $\varepsilon \in \{0, 1\}^{\text{Ed}(I)}$, $a_{I,\varepsilon} \in \mathcal{A}_n$ is the product of $\{a_i : i \in I\}$, each used once, in any linear order for which $\varepsilon_s = 0$ means a_s is applied before a_{s+1} and $\varepsilon_s = 1$ means a_{s+1} is applied before a_s ([2, **def:aIeps**]). This is well defined, and $\varepsilon \mapsto a_{I,\varepsilon}$ is a bijection onto the monomials of $\mathcal{A}_n$ with support I using each generator once. Postnikov's elements are

$$\mathbf{h}_m = \sum_{|I|=m, I \text{ proper}} a_{I,0}, \quad \mathbf{e}_b = \sum_{|I|=b, I \text{ proper}} a_{I,1}, \quad \mathbf{h}_0 = \mathbf{e}_0 = 1, \quad (2)$$

and $\mathcal{R}_e(t) = \sum_{|J|=e, r(J)=1} \sum_{\varepsilon} t^{|\varepsilon|} a_{J,\varepsilon}$ is the sum over *intervals* ([2, **eq:Rdef**]). Finally we recall [2, **lem:run**]: on a run $J = [i, i + \ell - 1]$ with orientation ε , the monomial $a_{J,\varepsilon}$ is nonzero on the bead state S if and only if

$$i \in S, \quad i + \ell \notin S, \quad \text{and} \quad i + s \in S \iff \varepsilon_s = 1 \quad (1 \leq s \leq \ell - 1), \quad (3)$$

in which case it moves the bead at i to $i + \ell$ and collects $q^{[n-1 \in J]}$.

Lemma 1 (runs act independently). *Let I be proper with runs $J_1, \dots, J_r$, $J_a = [i_a, i_a + \ell_a - 1]$. The closed intervals $\bar{J}_a := [i_a, i_a + \ell_a]$ are pairwise disjoint. Consequently $a_{I,\varepsilon} = \prod_a a_{J_a, \varepsilon|_{J_a}}$ in any order, and it is nonzero on S iff (3) holds for every run simultaneously.*

Proof. By maximality $i_a + \ell_a \notin I$, so the next site of I after $i_a + \ell_a - 1$ is at least $i_a + \ell_a + 1$; hence the next run starts at $\geq i_a + \ell_a + 1$ and $\bar{J}_a \cap \bar{J}_{a'} = \emptyset$. Generators indexed in different runs are non-adjacent, hence commute; and a_s for $s \in J_a$ only reads and writes sites in $\bar{J}_a$. $\square$

2 The two lemmas

Everything rests on the following pair. Neither is about Young diagrams: they are statements about occupancy and about the trace monoid.

Lemma 2 (a horizontal monomial empties its support; a vertical monomial fills its own). *Let I be proper and S a bead state.*

- (a) *If $a_{I,0} \sigma_S \neq 0$ and U is the resulting bead state, then $I \cap U = \emptyset$.*
- (b) *If $a_{I,1} \sigma_S \neq 0$ then $I \subseteq S$.*

Proof. Both are read off (3) run by run, legitimately by Lemma 1.

(a) For a run $J = [i, i + \ell - 1]$ with $\varepsilon = \mathbf{0}$, (3) says $i \in S$ and $i + s \notin S$ for $1 \leq s \leq \ell$. The move deletes the bead at i and creates one at $i + \ell$, so in U the sites $i, i + 1, \dots, i + \ell - 1$ are all empty — that is, $J \cap U = \emptyset$. Other runs do not touch $\bar{J}$. Taking the union over runs gives $I \cap U = \emptyset$.

(b) For a run $J = [i, i + \ell - 1]$ with $\varepsilon = \mathbf{1}$, (3) says $i \in S$ and $i + s \in S$ for $1 \leq s \leq \ell - 1$, i.e. $J \subseteq S$. $\square$

Corollary 3 (no repeated generator survives). *If $I' \cap I'' \neq \emptyset$ then $a_{I', \mathbf{0}} a_{I'', \mathbf{1}} = 0$ as an operator on $\bigoplus_k \text{QH}^*(\text{Gr}_{kn})$.*

Proof. Suppose $a_{I', \mathbf{0}} a_{I'', \mathbf{1}} \sigma_S \neq 0$ for some S (recall $a_{I', \mathbf{0}}$ acts first). Let U be the state after the first factor. By Lemma 2(a), $I' \cap U = \emptyset$; by Lemma 2(b) applied at U , $I'' \subseteq U$. Hence $I' \cap I'' \subseteq I' \cap U = \emptyset$. $\square$

Lemma 4 (a disjoint split names an orientation, and only the run-ends are free). *Let I be proper and let $I = I' \sqcup I''$. Then*

$$a_{I', \mathbf{0}} a_{I'', \mathbf{1}} = a_{I, \varepsilon(I'')} \quad \text{in } \mathcal{A}_n, \quad \text{where } \varepsilon(I'')_s = \begin{cases} 1, & s \in I'', \\ 0, & s \in I', \end{cases}$$

for every edge index $s \in \text{Ed}(I)$. Conversely, for each $\varepsilon \in \{0, 1\}^{\text{Ed}(I)}$ the splits $I = I' \sqcup I''$ with $\varepsilon(I'') = \varepsilon$ are exactly those obtained by putting $s \in I''$ for every edge index s with $\varepsilon_s = 1$, $s \in I'$ for every edge index s with $\varepsilon_s = 0$, and assigning each of the $r(I)$ sites of $L(I)$ freely to I' or to I'' . Hence

$$\sum_{I=I' \sqcup I''} t^{|I''|} a_{I', \mathbf{0}} a_{I'', \mathbf{1}} = \sum_{\varepsilon} t^{|\varepsilon|} (1+t)^{r(I)} a_{I, \varepsilon}. \quad (4)$$

Proof. The concatenated word w uses each a_s , $s \in I$, exactly once, so it is a representative of $a_{I, \varepsilon}$ for exactly one $\varepsilon \in \{0, 1\}^{\text{Ed}(I)}$ — provided we check that any two linear orders agreeing on all edges of I give the same element of $\mathcal{A}_n$. They do: two linear extensions of the partial order generated by the edge constraints differ by transpositions of adjacent-in-word incomparable elements, and two incomparable elements of I are non-adjacent in $\mathbb{Z}/n\mathbb{Z}$ (if $u, u+1$ both lie in I then $\{u, u+1\}$ is an edge of I and they are comparable), hence commute.

It remains to identify ε . Fix an edge $\{s, s+1\}$ of I and consider the four cases. If $s \in I'$ and $s+1 \in I'$, then $s, s+1$ lie in the same run of I' (they are adjacent and both in I'), and $\mathbf{0}$ puts a_s before a_{s+1} : $\varepsilon_s = 0$. If $s \in I'$ and $s+1 \in I''$, then a_s is in the first factor and a_{s+1} in the second, so a_s is applied first: $\varepsilon_s = 0$. If $s \in I''$ and $s+1 \in I'$, then a_{s+1} is in the first factor and a_s in the second, so a_{s+1} is applied first: $\varepsilon_s = 1$. If $s \in I''$ and $s+1 \in I''$, they lie in the same run of I'' and $\mathbf{1}$ puts a_{s+1} before a_s : $\varepsilon_s = 1$. In all four cases $\varepsilon_s = [s \in I'']$, which is the assertion.

The converse is immediate: the map $I'' \mapsto \varepsilon(I'')$ reads off the membership of the $|I| - r(I)$ edge-indexing sites and forgets the membership of the $r(I)$ sites of $L(I)$, which by (1) is the whole of the remaining data. Each fibre therefore has $2^{r(I)}$ elements, one for each subset $L'' \subseteq L(I)$, and $|I''| = |\varepsilon| + |L''|$. Summing $t^{|I''|}$ over the fibre gives $t^{|\varepsilon|} (1+t)^{r(I)}$, which is (4). $\square$

Remark 5 (what happened to the classical proof). Lemma 4 is the cylindric replacement for three lemmas of [3]: “a 2×2 kills every splitting”, “splittings factor over components”, and “a ribbon has exactly two splittings”. All three are absorbed. The 2×2 obstruction is invisible here because a bead move of the form (3) can never contain one — the monomials of $\mathcal{A}_n$ with a given support are the 2×2 -free shapes, by [2, **thm:A**]; the factorisation over components is Lemma 1; and “exactly two splittings” is the free choice of the single site $L(J)$ of a one-run J . In the classical argument the

factor $(1+t)$ emerges after a page of interlacing inequalities ([3, lem:ribbon]) as $\alpha_1 \in \{a_1 - 1, a_1\}$; here it is the last generator of the run, which may be applied in either factor because no constraint mentions it. The two are the same fact: α_1 is the number of cells of the first row placed in the horizontal half, and the run-end site is the corresponding generator.

3 The splitting identity and its derivative

Theorem 6 (cylindric splitting identity). *For $1 \leq e \leq n-1$, as operators on $\bigoplus_k \text{QH}^*(\text{Gr}_{kn})$,*

$$\sum_{b=0}^e t^b \mathbf{h}_{e-b} \mathbf{e}_b = \sum_{\substack{I \subseteq \mathbb{Z}/n\mathbb{Z} \\ |I|=e}} \sum_{\varepsilon \in \{0,1\}^{\text{Ed}(I)}} t^{|\varepsilon|} (1+t)^{r(I)} a_{I,\varepsilon}. \quad (5)$$

Proof. Since $e \leq n-1$, every I with $|I| = e$ is proper, and so is every subset of it; thus all the monomials occurring below are among those of (2). Expanding by (2),

$$\sum_{b=0}^e t^b \mathbf{h}_{e-b} \mathbf{e}_b = \sum_{b=0}^e t^b \sum_{|I'|=e-b} \sum_{|I''|=b} a_{I',0} a_{I'',1}.$$

By Corollary 3 every term with $I' \cap I'' \neq \emptyset$ vanishes, so the double sum is over ordered pairs of *disjoint* sets with $|I' \sqcup I''| = e$. Grouping these by $I := I' \sqcup I''$ and applying (4) to each I gives (5). $\square$

Corollary 7 ($t = -1$ recovers Postnikov's vanishing). *For $1 \leq e \leq n-1$, $\sum_{b=0}^e (-1)^b \mathbf{h}_{e-b} \mathbf{e}_b = 0$; that is, the coefficients of $t^1, \dots, t^{n-1}$ in $(1 + \sum_i \mathbf{e}_i t^i)(1 + \sum_j \mathbf{h}_j (-t)^j)$ vanish.*

Proof. $r(I) \geq 1$ for $|I| = e \geq 1$, so every term on the right of (5) carries a factor $(1+t)$. $\square$

This is the part of Postnikov's Lemma (lem:e-h-commute-in-AN, [1, ll.2882–2893]) that lives below degree n ; it is recovered here, not assumed. The new content is one derivative further.

Theorem 8 (closed formula for the ribbon adder at $t = -1$). *For $1 \leq e \leq n-1$, as operators on $\bigoplus_k \text{QH}^*(\text{Gr}_{kn})$,*

$$\boxed{\mathcal{R}_e(-1) = \sum_{b=1}^e (-1)^{b-1} b \mathbf{h}_{e-b} \mathbf{e}_b.} \quad (6)$$

Proof. Both sides of (5) are polynomials in t with operator coefficients; differentiate in t and set $t = -1$.

On the right, the I -th summand is $\sum_{\varepsilon} t^{|\varepsilon|} (1+t)^{r(I)} a_{I,\varepsilon}$, whose derivative is $\sum_{\varepsilon} (|\varepsilon| t^{|\varepsilon|-1} (1+t)^{r(I)} + r(I) t^{|\varepsilon|} (1+t)^{r(I)-1}) a_{I,\varepsilon}$. At $t = -1$ the first term vanishes because $r(I) \geq 1$, and the second vanishes unless $r(I) = 1$, in which case it equals $\sum_{\varepsilon} (-1)^{|\varepsilon|} a_{I,\varepsilon}$. Now $|I| = e$ and $r(I) = 1$ says exactly that I is an interval of length e , so the surviving sum is $\sum_{|J|=e, r(J)=1} \sum_{\varepsilon} (-1)^{|\varepsilon|} a_{J,\varepsilon} = \mathcal{R}_e(-1)$ by [2, thm:A].

On the left, $\frac{d}{dt} \sum_b t^b \mathbf{h}_{e-b} \mathbf{e}_b = \sum_{b \geq 1} b t^{b-1} \mathbf{h}_{e-b} \mathbf{e}_b$, which at $t = -1$ is $\sum_{b \geq 1} (-1)^{b-1} b \mathbf{h}_{e-b} \mathbf{e}_b$. $\square$

Remark 9. The same derivative, performed in Λ rather than in $\mathcal{A}_n$, is the classical identity $p_e = \sum_{b=1}^e (-1)^{b-1} b h_{e-b} e_b$: from $H(z)E(-z) = 1$ one gets $\frac{\partial}{\partial t} H(z)E(tz)|_{t=-1} = zE'(-z)/E(-z) = z \frac{d}{dz} \log H(z) = \sum_{e \geq 1} p_e z^e$. So (6) is literally the statement that the classical expression of p_e in the h 's and e 's is valid verbatim on the cylinder in degrees $\leq n-1$ — which is why no wrap term can appear there. The wrap is confined to $e = n$, where $I = \mathbb{Z}/n\mathbb{Z}$ is not proper and Theorem 6 has nothing to say.

4 The cylindric Newton identity

Lemma 10 (the strip adders are the quantum multiplications, in the full range). *Fix $1 \leq k \leq n-1$ and write M_x for quantum multiplication by $x \in \text{QH}^*(\text{Gr}_{kn})$. Then on $\text{QH}^*(\text{Gr}_{kn})$,*

$$\mathbf{h}_j = M_{\sigma_j} \quad (0 \leq j \leq n-1), \quad \mathbf{e}_b = M_{\sigma_{1b}} \quad (0 \leq b \leq n-1),$$

with the convention $\sigma_j = 0$ for $j > n-k$ and $\sigma_{1b} = 0$ for $b > k$.

Proof. For $1 \leq j \leq n-k$ and $1 \leq b \leq k$ this is Postnikov's quantum Pieri corollary [1, cor:pieri-AN, ll.2935–2947], read at source; the cases $j = 0, b = 0$ are trivial. It remains to see that the operators vanish outside those ranges, which the bead model makes immediate.

If $b > k$: by Lemma 2(b) a nonzero $a_{I,1}$ with $|I| = b$ requires $I \subseteq S$, impossible since $|S| = k < b$. Hence $\mathbf{e}_b = 0 = M_{\sigma_{1b}}$.

If $j > n-k$: let I be proper with $|I| = j$ and $a_{I,0}\sigma_S \neq 0$. For each run $J_a = [i_a, i_a + \ell_a - 1]$, (3) with $\varepsilon = \mathbf{0}$ forces the ℓ_a sites $i_a + 1, \dots, i_a + \ell_a$ to be empty in S . These sets are pairwise disjoint over the runs, being contained in the pairwise disjoint $\bar{J}_a$ of Lemma 1 and each of size ℓ_a ; so S has at least $\sum_a \ell_a = j$ empty sites. But S has exactly $n-k < j$ empty sites. Hence $\mathbf{h}_j = 0 = M_{\sigma_j}$. $\square$

Theorem 11 (Newton, and the defect). *Let $1 \leq k \leq n-1$ and put $p_e := \mathcal{R}_e(-1)$ for $1 \leq e \leq n-1$. Then, as operators on $\text{QH}^*(\text{Gr}_{kn})$:*

(i) *each p_e is a quantum multiplication operator, namely $p_e = M_{\varphi(p_e)}$, where $\varphi : \Lambda[q] \rightarrow \text{QH}^*(\text{Gr}_{kn})$ is the ring homomorphism with $\varphi(e_i) = \sigma_{1i}$; in particular the p_e commute with one another and with every $\mathbf{h}_j$;*

(ii) $r \mathbf{h}_r = \sum_{e=1}^r p_e \mathbf{h}_{r-e}$ for $1 \leq r \leq n-1$;

(iii) $\sum_{e=1}^{n-1} p_e \mathbf{h}_{n-e} = (-1)^{k-1} (n-k) q \cdot \text{Id}$.

Proof. φ is well defined because $\Lambda = \mathbb{Z}[e_1, e_2, \dots]$ is a free polynomial ring, and we extend it $\mathbb{Z}[q]$ -linearly. We first record

$$\varphi(h_j) = \sigma_j \quad (0 \leq j \leq n-1). \quad (7)$$

Indeed $\sum_{b=0}^j (-1)^b h_{j-b} e_b = 0$ in Λ for $j \geq 1$; applying φ gives $\sum_{b=0}^j (-1)^b \varphi(h_{j-b}) \sigma_{1b} = 0$. On the other side, the coefficient of t^j in Postnikov's presentation $e(t)h(-t) = 1 + (-1)^{n-k} q t^n$ [1, eq:QH, ll.500–507] is $\sum_{b=0}^j (-1)^{j-b} \sigma_{1b} \sigma_{j-b} = 0$ for $1 \leq j \leq n-1$, i.e. the same recursion. Both sequences start at $\varphi(h_0) = 1 = \sigma_0$, and the recursion determines the j -th term from the earlier ones (the $b = 0$ term is $\varphi(h_j)$ resp. σ_j , with coefficient ± 1), so (7) follows by induction.

(i) By Theorem 8 and Lemma 10,

$$p_e = \sum_{b=1}^e (-1)^{b-1} b \mathbf{h}_{e-b} \mathbf{e}_b = \sum_{b=1}^e (-1)^{b-1} b M_{\sigma_{e-b}} M_{\sigma_{1b}} = M_{\sum_{b=1}^e (-1)^{b-1} b \sigma_{e-b} * \sigma_{1b}},$$

using that $\text{QH}^*(\text{Gr}_{kn})$ is a commutative associative ring, so $M_x M_y = M_{x*y}$. By (7) and the classical identity $p_e = \sum_{b \geq 1} (-1)^{b-1} b h_{e-b} e_b$ in Λ (Remark after Theorem 8), the subscript is $\varphi(p_e)$. Multiplication operators of a commutative ring commute with one another, and $\mathbf{h}_j = M_{\sigma_j}$.

(ii) For $1 \leq r \leq n-1$, by (i), Lemma 10 and (7),

$$\sum_{e=1}^r p_e \mathbf{h}_{r-e} = \sum_{e=1}^r M_{\varphi(p_e)} M_{\varphi(h_{r-e})} = M_{\varphi(\sum_{e=1}^r p_e h_{r-e})} = M_{\varphi(r h_r)} = r M_{\sigma_r} = r \mathbf{h}_r,$$

the third equality being Newton's identity in Λ .

(iii) The same computation at $r = n$, where $\mathbf{h}_{n-e} = M_{\varphi(h_{n-e})}$ still holds for $1 \leq e \leq n-1$ because then $n-e \leq n-1$, gives

$$\sum_{e=1}^{n-1} p_e \mathbf{h}_{n-e} = M_{\varphi(\sum_{e=1}^{n-1} p_e h_{n-e})} = M_{\varphi(n h_n - p_n)} = M_{n\varphi(h_n) - \varphi(p_n)}.$$

It remains to evaluate the two scalars. Taking $j = n$ in the Λ -recursion, $\varphi(h_n) = -\sum_{b=1}^n (-1)^b \sigma_{1^b} \varphi(h_{n-b}) = -\sum_{b=1}^n (-1)^b \sigma_{1^b} \sigma_{n-b}$ by (7); and the t^n coefficient of $e(t)h(-t) = 1 + (-1)^{n-k} q t^n$ reads $\sum_{b=0}^n (-1)^{n-b} \sigma_{1^b} \sigma_{n-b} = (-1)^{n-k} q$, whose $b=0$ term is $\sigma_n = 0$ (as $n > n-k$); multiplying by $(-1)^n$ gives $\sum_{b=1}^n (-1)^b \sigma_{1^b} \sigma_{n-b} = (-1)^k q$. Hence $\varphi(h_n) = (-1)^{k-1} q$.

For $\varphi(p_n) = \sum_{b=1}^n (-1)^{b-1} b \varphi(h_{n-b}) \sigma_{1^b}$, the factor σ_{1^b} vanishes for $b > k$ and, for $1 \leq b \leq n-1$, $\varphi(h_{n-b}) = \sigma_{n-b}$ vanishes unless $n-b \leq n-k$, i.e. $b \geq k$; the term $b = n$ carries $\sigma_{1^n} = 0$ since $n > k$. So only $b = k$ survives: $\varphi(p_n) = (-1)^{k-1} k \sigma_{n-k} * \sigma_{1^k} = (-1)^{k-1} k q$, using the single quantum relation $e_k h_{n-k} = q$ [2, `thm:C(ii)`]. Therefore

$$n\varphi(h_n) - \varphi(p_n) = n(-1)^{k-1} q - (-1)^{k-1} k q = (-1)^{k-1} (n-k) q. \quad \square$$

Corollary 12 (quantum Murnaghan–Nakayama, unconditionally). *For $1 \leq e \leq n-1$ and $\lambda \in P_{kn}$,*

$$\sum_{\mu: \mu/\lambda \text{ a cylindric } e\text{-ribbon}} (-1)^{\text{ht}(\mu/\lambda)} q^{w(\mu/\lambda)} \sigma_\mu = \varphi(p_e) * \sigma_\lambda,$$

where w counts seam crossings. In particular $[\mathcal{R}_e(-1), \mathbf{h}_j] = 0$ for all e, j .

Proof. The left side is $\mathcal{R}_e(-1) \sigma_\lambda$ by [2, `eq:ribbondef` and `thm:A`]; apply Theorem 11(i). $\square$

This is [2, `prop:MN`], whose proof there assumed Theorem `thm:B(ii)`; the dependence is now reversed and the assumption discharged.

5 The boundary cases $k = 1$ and $k = n-1$ (gap (G3))

Proposition 13. *Let $1 \leq e \leq n-1$.*

(a) *On $\text{QH}^*(\text{Gr}_{1,n})$, $\mathcal{R}_e(t) = \mathcal{R}_e(0)$: the operator does not depend on t at all.*

(b) *On $\text{QH}^*(\text{Gr}_{n-1,n})$, $\mathcal{R}_e(t) = t^{e-1} \mathcal{R}_e(1)$: the operator is t^{e-1} times a t -free operator.*

Consequently, at $k \in \{1, n-1\}$ the identity $[\mathcal{R}_e(t), \mathbf{h}_j] = 0$ holds either for every t or for no t , and it does hold for every t by Corollary 12. The uniqueness assertion of [2, `thm:B(i)`] is therefore not merely untested at the boundary: it is vacuous there, because the t -line has collapsed to a single monomial. This is a rank degeneracy of the Hall–Littlewood grading, not a statement about $\mathbb{P}^{n-1}$.

Proof. Recall from (3) (equivalently [2, eq:ribbondef]) that the (S, j) term of $\mathcal{R}_e(t)$ carries $t^{\#\{1 \leq s \leq e-1: j+s \in S\}}$.

(a) $|S| = 1$, say $S = \{\beta\}$, and the only legal source is $j = \beta$. For $1 \leq s \leq e-1 \leq n-2$ we have $\beta + s \not\equiv \beta$, so $\beta + s \notin S$ and the exponent is 0.

(b) $|S| = n-1$; let g be the unique empty site. A move $j \mapsto j+e$ is legal iff $j+e \notin S$, i.e. $j+e = g$. For $1 \leq s \leq e-1$, $j+s \equiv g = j+e$ would force $s \equiv e \pmod{n}$, impossible; so $j+s \neq g$, i.e. $j+s \in S$, and the exponent is $e-1$ for every legal term. $\square$

Remark 14. (a) and (b) are exchanged by the duality ρ of [2, cor:duality], $\rho \mathcal{R}_e(t) \rho^{-1} = t^{e-1} \mathcal{R}_e(1/t)$, which is consistent: a t -free operator is carried to a t^{e-1} -multiple of a t -free one.

6 Computational verification

All checks are exact and symbolic (SymPy over $\mathbb{Z}[q, t]$), in `proofs/code-2026-09-24/`, built on the $\mathcal{A}_n$ model of `proofs/code-q237/anTL.py`, which implements a_i as the bead move, so that the nil-Temperley–Lieb relations are consequences of the model rather than inputs to it. Each check below tests a *step* of the proof, not its conclusion; the conclusion (Theorem 11(ii)–(iii)) was already verified for $3 \leq n \leq 7$ in [2] and was deliberately not re-run.

1. **Corollary 3** (`check1_no_repeat.py`). For $2 \leq n \leq 6$, all $0 \leq k \leq n$, and all ordered pairs of nonempty proper I', I'' with $I' \cap I'' \neq \emptyset$: $a_{I',0} a_{I'',1} = 0$. 27 846 pairs, **0 violations**.
2. **Lemma 4** (`check4_split_orientation.py`). For $2 \leq n \leq 7$, all k , all proper I and all splits $I = I' \sqcup I''$: $a_{I',0} a_{I'',1} = a_{I,\varepsilon}$ with $\varepsilon_s = [s \in I'']$. 22 776 splits, **0 violations**.
3. **Theorem 6** (`check2_C1_identity.py`). For $2 \leq n \leq 7$, all $0 \leq k \leq n$, all $1 \leq e \leq n-1$: the two sides of (5) agree entrywise as polynomials in $\mathbb{Z}[q, t]$. **0 violations**.
4. **Theorem 8** (`check3_pe_formula.py`). Same ranges: $\sum_{b \geq 1} (-1)^{b-1} b \mathbf{h}_{e-b} \mathbf{e}_b$ equals the independent bead definition of $\mathcal{R}_e(t)$ specialised at $t = -1$. **0 violations**.
5. **Lemma 10, vanishing half** (`check5_vanishing.py`). For $2 \leq n \leq 8$, all k : $\mathbf{e}_b = 0$ for $b > k$ and $\mathbf{h}_j = 0$ for $j > n-k$. **0 violations**.
6. **Proposition 13** (`g3_boundary.py`). For $3 \leq n \leq 8$ and $k \in \{1, n-1\}$, all $1 \leq e \leq n-1$: the set of t -degrees occurring in the matrix of $\mathcal{R}_e(t)$ is $\{0\}$ when $k = 1$ and $\{e-1\}$ when $k = n-1$.

7 Gaps

(H1) **Uniqueness of $t = -1$ is still open.** We prove $[\mathcal{R}_e(-1), \mathbf{h}_j] = 0$ (Corollary 12); we do *not* prove that $t = -1$ is the only such point for $2 \leq k \leq n-2$, which is the remaining half of [2, thm:B(i)] and is verified only for $4 \leq n \leq 6$. Theorem 6 does not settle it: it shows that $\sum_{c \geq 1} (1+t)^c A_c(t)$ is a multiplication operator for *every* t , where $A_c(t)$ collects the I with $r(I) = c$, and one cannot isolate $A_1 = \mathcal{R}_e$ from that. Proposition 13 shows the question is vacuous at $k \in \{1, n-1\}$, which is why the hypothesis $2 \leq k \leq n-2$ is the right one, but it does not prove uniqueness inside that range.

- (H2) **The route predicted by the brief was not taken, and its prediction is untested.** The WAKE brief for this session conjectured that the defect at $r = n$ would be exhibited by enumerating wrap-around contributions indexed by the $n - k$ gaps of the bead configuration, and proposed that as a falsifiable test. The proof above never enumerates wrap contributions: the $n - k$ arises as $n \cdot 1 - k \cdot 1$, from $n\varphi(h_n)$ against the single surviving index $b = k$ in $\varphi(p_n)$. The gap-indexed picture may still be true; *it is neither used nor checked here*, and nothing in this note is evidence for it.
- (H3) **Lemma 10 rests on a reading of Postnikov.** `cor:pieri-AN` is stated at [1, ll.2935–2947] for $i = 1, \dots, k$ and $j = 1, \dots, n - k$ only; the extension to the full range $0 \leq \cdot \leq n - 1$ is the vanishing argument given here, which is ours. The line numbers are internal locators in the e-print I hold, not published theorem numbers. The application-order convention of [2, `rem:conv`] (gap (G4) there) remains a reading of ll.2839–2877, not a quotation, and Lemma 4 is entirely a statement about application order — if that reading is wrong, the four-case analysis in its proof inverts and $\varepsilon_s = [s \in I']$ instead. The identity (5) would then hold with **h** and **e** exchanged, which is the $t \mapsto 1/t$ image of the same statement; nothing downstream changes.
- (H4) **Novelty unchecked.** See the status box. In particular I have not read Bertram–Ciocan-Fontanine–Fulton or Morrison–Sottile, and a quantum Murnaghan–Nakayama rule for $\mathrm{QH}^*(\mathrm{Gr}_{kn})$ may be known; if so, Corollary 12 is a second proof of a known rule and must be described that way. No sentence here claims priority.
- (H5) **Gap (G2) of [2] is moot, not resolved.** That gap proposed proving the $q = 0$ case first and lifting. The present proof is uniform in q and never performs that reduction, so the reduction remains uninspected. It is no longer on the critical path.

References

- [1] A. Postnikov, *Affine approach to quantum Schubert calculus*, `arXiv:math/0205165v2`. Cited by internal L^AT_EX labels and line numbers of the e-print source at `sources/math0205165-postnikov/postnikov-affine-v2.tex`, read at source on 2026-09-23 and 2026-09-24.
- [2] Clio, *The height of a ribbon is the orientation of a heap: the Hall–Littlewood t -family inside the affine nil-Temperley–Lieb algebra*, `proofs/2026-09-23-c2-ribbon-height-is-a-heap-orientation.tex`, `commit clio-vega/proofs@0a01bb9`.
- [3] Clio, *The multiplication defect of the Hall–Littlewood ribbon family*, `proofs/2026-09-03-Q75-multiplication-defect.tex`.